

4.1

$$a. N=20 \quad \sum y_i^2 = 7825.94 \quad \bar{y} = 19.21 \quad SSR = 375.47$$

$$\Rightarrow R^2 = \frac{SSR}{SSTO} = \frac{375.47}{445.458} = 0.8429$$

$$b. R^2 = 0.7911, \quad SST = 725.94, \quad N=20$$

$$\Rightarrow \hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{\sum (y_i - \hat{y}_i)^2}{N-2} = \frac{SSE}{N-2} = \frac{151.6489}{20-2} = 8.4249$$

$$R^2 = 0.7911 = 1 - \frac{SSE}{SST} = 1 - \frac{SSE}{725.94} \Rightarrow SSE = 151.6489$$

$$c. \sum (y_i - \bar{y})^2 = 631.63, \quad \sum \hat{e}_i^2 = 182.85$$

$$\Rightarrow R^2 = \frac{SSR}{SSTO} = \frac{631.63 - 182.85}{631.63} = 0.7105$$

$$4.3 \quad a. \quad \hat{y} = 1.2 + 0.8x \Rightarrow \hat{y}_0 = 1.2 + 0.8 \times 4 = 4.4$$

$$b. \hat{var}(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right] = 1.2 \times \left(1 + \frac{1}{5} + \frac{9}{10} \right) = 2.52$$

$$\hookrightarrow \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$$

$$se(f) = \sqrt{\hat{var}(f)} = 1.5875$$

$$c. 95\% \text{ P.I.} \Rightarrow \hat{y}_0 \pm t_c se(f) \Rightarrow 4.4 \pm 3.1824 \times 1.5875 = [-0.652, 9.452]$$

$$d. 99\% \text{ P.I.} \Rightarrow \hat{y}_0 \pm t_c se(f) \Rightarrow 4.4 \pm 5.8409 \times 1.5875 = [-4.872, 13.672]$$

$$e. x = \bar{x} = 1$$

$$\hat{y}_0 = 1.2 + 0.8 \times 1 = 2$$

$$\hat{var}(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right] = 1.2 \times \left(1 + \frac{1}{5} + 0 \right) = 1.44$$

width of interval for
part c = 10.1

$$se(f) = \sqrt{\hat{var}(f)} = 1.2$$

width of interval for
part e = 7.64

$$95\% \text{ P.I.} \Rightarrow \hat{y}_0 \pm t_c se(f) \Rightarrow 2 \pm 3.1824 \times 1.2 = [-1.819, 5.819]$$

part e interval $\Rightarrow$ narrower, se of forecast is minimized
when prediction point (x_0) is $\bar{x}$